

Rahul Rawat

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PROFILE

Master of Science Data Science student at SRH Heidelberg who enjoys turning real, incomplete and inconsistent data into reliable, reusable solutions. I build Python tools that analyse, clean, transform and connect heterogeneous data sources, systematically check data quality, and feed trustworthy data into machine learning and decision models. I work with pandas, NumPy and scikit learn and care about writing maintainable code, which is exactly the profile this working student role at alloqis is looking for.

EDUCATION

M.Sc. Data Science and Analytics, SRH University of Applied Sciences Heidelberg

Apr 2025 to Present

B.Sc. Computer Science, GL Bajaj Institute of Technology and Management

2019 to 2023

EXPERIENCE

eRay GmbH

Data Scientist · Oct 2025 to Mar 2026

- Built an end to end recursive time series pipeline in Python that forecasts four water quality indicators from real, imperfect sensor data over a six month industry collaboration.
- Reconstructed missing winter readings with MICE imputation and built standardised checks so the pipeline supplies decision models with reliable, dependable data rather than gaps.
- Wrapped everything in an orchestrator with gate checks and ecological bounds that halts on failed imputation rather than letting bad data cascade, exactly the systematic data quality approach this role describes.
- Benchmarked six models with scikit learn and gradient boosting libraries and used CatBoost multi quantile regression for asymmetric 80 percent prediction intervals.

PERSONAL PROJECTS

Movie Analytics and Machine Learning Pipeline: Python, SQL, GCP BigQuery, BigQuery ML, Looker Studio

- Engineered a Silver data layer for trustworthy analytics with schema enforcement, safe type casting and deduplication via window functions, a direct example of systematic data quality assurance.
- Built an end to end pipeline through a Bronze, Silver and Gold medallion architecture, fully automated with no manual steps and version controlled for reproducibility.
- Trained a machine learning classifier on the cleaned data, deliberately splitting features across two tables to prevent data leakage.

Real Time Flight Tracking Data Pipeline: Python, PySpark, BigQuery, dbt, Airflow, GCP

- Cleaned, transformed and connected real world data from four heterogeneous sources, airport, aircraft, weather and live position feeds, into analysis ready tables.
- Computed each aircraft's nearest airport along the way and shaped the data with dbt into reusable, well structured tables.
- Built reproducible Python and PySpark components that handle messy live data at scale on Google Cloud.

CreditIQ, Fairness by Design Credit Scoring System: Python, scikit learn, AIF360, SHAP

- Built a maintainable machine learning pipeline in Python, backed by unit tests at 100 percent branch coverage, that trains, evaluates and explains a scoring model.
- Lifted the model's Disparate Impact ratio from a failing 0.79 to a compliant 0.88 and cut the false negative rate from 44 percent to 16.7 percent.
- Used SHAP to make model behaviour understandable and traceable for a non technical audience.

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning: Python, scikit learn, Pandas, Seaborn

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of 768 records with 10 fold cross validation.
- Caught biologically impossible zero values the original authors had overlooked, then applied IQR based outlier removal and proper imputation on the inconsistent data.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, then wrote up the findings as an IEEE style paper with an honest limitations section.

LANGUAGES

- English: fluent, professional working proficiency
- German: B1, in progress